HOPE OGBONS

Senior Agentic / Forward-Deployed AI Engineer

Applied LLM Systems • Autonomous Agents • Production AI Infrastructure

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PROFESSIONAL SUMMARY

Forward-Deployed AI Engineer specializing in **production-grade agentic systems**, deterministic LLM orchestration, and real-world backend integration. Proven track record delivering **secure, observable, cost-optimized AI systems** across finance, healthcare, customer support, travel, and multi-agent environments. Strong focus on **hallucination mitigation, system reliability, and stakeholder-facing rapid iteration**.

KEY SKILLS

- **Agentic AI**
  Autonomous agents, tool execution engines, state machines, multi-agent orchestration
- **LLMs & Runtimes**
  OpenAI (GPT-4o, GPT-5), AWS Bedrock, MCP, JSON Schema validation
- **Reliability & Safety**
  Guardrails, Pydantic, deterministic execution, observability, audit trails
- **Backend & Infra**
  Python, FastAPI, AWS Lambda, App Runner, SQS, Redis, Terraform
- **Databases & Storage**
  Postgres, DynamoDB, MongoDB, Redis, Vector DBs (Pinecone, Weaviate, pgvector),
  Search (OpenSearch), Graph (Neo4j), Object Storage (S3)
- **RAG & Data**
  Cost-optimized RAG, S3 vectors, HuggingFace embeddings
- **Full Stack**
  Next.js, React, NodeJS, Python, Gradio, Langfuse, OpenTelemetry

PROFESSIONAL EXPERIENCE

Co-Founder / Forward-Deployed AI Engineer

RoboOffice, *Lagos, Nigeria*
(May 2023 – Present)

RoboCare Intelligence Platform (Backend): An AI-Powered Care Matching & Orchestration System.
MyWoosah Inc, Georgia, USA
https://github.com/hopeogbons/robocare_backend

- **Context-Aware AI Matching Engine:**

    - ***Core Systems:*** OpenAI GPT-4o-mini (RAG), pgvector (Vector Search), Hybrid SQL/Semantic Scoring Algorithm.
    - ***Impact:*** Architected a high-fidelity matching system that combines rigid compliance filters (verified certifications, availability) with semantic analysis of care needs. Implemented a custom scoring algorithm that weights verified credentials (2x bonus) and strictly enforces schedule compatibility, eliminating irrelevant candidate suggestions.
- **Multi-Tenant Workflow Orchestration:**
    - ***Core Systems:*** Django Multi-Tenancy (Row-Level Security pattern), Atomic Service Scheduling, Complex State Machines.
    - ***Impact:*** Built a robust SaaS infrastructure supporting complete data isolation for B2B2C clients. Designed an atomic ServiceSchedule system that handles complex recurring patterns (daily/weekly/monthly) independently of the parent service, ensuring accurate billing and conflict-free roster management.
- **Persona-Based Communication & Notification Layer:**
    - ***Core Systems:*** Django Channels (WebSockets), Agora (Video SDK), Persona-Scoped Routing Logic.
    - ***Impact:*** Developed a unified communication hub that intelligently routes real-time alerts to specific user personas (e.g., 'Employee' vs 'Family' roles). Integrated seamless in-app video interviewing that streamlines the recruitment lifecycle without external tools.
- **Resilient Background Processing & Financials:**
    - ***Core Systems***: Celery, Redis, Stripe Connect, Tenant-Aware Task Queues.
    - ***Impact***: Engineered a fail-safe background processing system that respects tenant boundaries for critical financial operations. Implemented idempotent hour-balance renewal and payment retry logic, preventing revenue leakage and ensuring 100% transactional integrity.

AI Digital Twin: A Context-Adaptive Persona Engine for High-Fidelity Professional Representation.
Andela, New York, USA
https://github.com/hopeogbons/digital-twin

- **Serverless Generative AI Microservice:**
    - ***Core Systems:*** FastAPI, AWS Lambda, AWS Bedrock (Nova Lite), Python 3.12, Boto3.
    - ***Impact:*** Built and scaled a low-latency conversational backend on a serverless architecture, integrating AWS Bedrock for cost-efficient inference. Engineered a lightweight adapter pattern to unify model interactions, ensuring zero specific infrastructure maintenance while handling burst traffic with minimal cold starts.
- **Production-Grade Infrastructure as Code:**
    - ***Core Systems:*** Terraform, AWS IAM, API Gateway v2, CloudFront, Route53.
    - ***Impact:*** Architected a complete, reproducible cloud environment from scratch, automating the provisioning of secure networking, compute, and storage resources. Reduced environment deployment time to under 10 minutes and enforced strict least-privilege IAM policies, eliminating manual configuration drift in production.
- **Stateful "Digital Twin" Persona Engine:**
    - ***Core Systems:*** AWS S3 (JSON Persistence), Custom Context Injection, Dynamic Prompt Engineering.
    - ***Impact:*** Engineered a persistent memory system that maintains multi-turn conversation state across stateless Lambda invocations. Implemented a robust context-injection strategy that dynamically loads personal facts and communication styles, driving high-fidelity user simulations and consistent long-term engagement.
- **Global Frontend Delivery System:**

- ○ ***Core Systems:*** Next.js 14, React, Tailwind CSS, AWS CloudFront, ACM (SSL/TLS).
- ○ ***Impact:*** Deployed a responsive, edge-cached frontend application optimized for global performance. Configured strict SSL/TLS security and custom domain routing via CloudFront and Route53, delivering sub-100ms content load times and 99.99% availability for end-users.

Agent-in-a-loop: Breaking Down Complex Goals via Autonomous Todo-List Generation.
Andela, *New York, USA*
https://github.com/hopeogbons/agent-in-a-loop

- **Custom Agentic Runtime & Orchestration Implementation:**
  - ***Core System:*** Python, OpenAI GPT-4o API, Raw JSON Schema Validation, Custom Tool Execution Engine.
  - ***Impact:*** Architected a lightweight, dependency-free agent production loop that bypassed heavy frameworks (like LangChain), reducing inference latency and giving engineering teams full determinism over tool execution sequences in safety-critical workflows.
- **State-Aware Planning & Self-Correction Module:**
  - ***Core* System:** In-Context State Management, Recursive Function Calling, Dynamic Context Injection.
  - ***Impact:*** Deployed a persistent "Todo-List" heuristic that allowed the agent to self-track progress across long-horizon tasks, effectively eliminating hallucinated steps for complex user requests and increasing successful task completion rates in multi-turn sessions.
- **Real-Time Observability & Developer Experience:**
  - ***Core System:*** Rich CLI Library, Structured Event Logging, Interactive Debugging Interfaces.
  - ***Impact:*** Built a high-visibility tracing system that exposed the model's "Chain of Thought" and tool outputs in real-time, significantly accelerating prompt iteration cycles with stakeholders and reducing mean-time-to-resolution for logic defects during pilot deployments.

AI Battle Tank: Deceptive Swarm & Multi-Agent Orchestration for the MCP Game
Andela, *New York, USA*
https://github.com/hopeogbons/ai-battle-tank

- **Adaptive Multi-Agent Negotiation System:**
  - ***Core Systems:*** Hierarchical LLM Architecture (Orchestrator vs. Specialist Agents), Hybrid SQL/NLP Trust Analysis, Asynchronous Event Loop (Asyncio).
  - ***Impact:*** Architected a self-correcting autonomous agent capable of forming and monitoring complex alliances in real-time competitive environments. Implemented a persistent "Trust & Betrayal" memory layer that analyzes historical interactions to detect opportunistic behavior, enabling the system to dynamically shift strategies (Defensive/Aggressive) 40% faster than heuristic baselines.
- **Adversarial Swarm & Deceptive Orchestration Grid:**
  - ***Core Systems:*** Distributed State Synchronization, Role-Based Swarm Coordination, Automated Social Engineering.
  - ***Impact:*** Engineered a high-concurrency "botnet" that coordinates multiple sleeper agents to maximize a central leader's score. Developed a "rotation-based support" protocol that covertly funnels resources while maintaining plausible deniability through generated cover messages, successfully demonstrating robustness against standard collusion-detection algorithms.
- **Real-Time Game Telemetry & Prompt Evolution Pipeline:**
  - **Core Systems:** SQLite Performance Tracking, A/B Testing Framework for Prompts, Automated Reporting.

    - **Impact:** Built a dedicated analytics engine to quantify agent performance across matches. Tracked specific KPIs (Trust Score, Betrayal Rate) to iteratively refine system prompts, resulting in a data-driven optimization loop that increased mutual alliance formation by 15% over initial baselines.

Alex: A Multi-Agent Financial Analysis Platform
Andela, *New York, USA*
https://github.com/hopeogbons/alex

- **Resilient Multi-Agent Orchestration:**
    - ***Core Systems:*** OpenAI Agents SDK, AWS Lambda, Amazon SQS, Tenacity (Retry Logic).
    - ***Impact:*** Engineered a decoupled, event-driven orchestration layer that coordinates five specialized AI agents to generate comprehensive financial reports. Implemented asynchronous state management via SQS and robust exponential backoff strategies, enabling massive parallelism that reduced complex portfolio analysis time by ~70% compared to synchronous sequential execution.
- **Autonomous Market Research Agent:**
    - ***Core Systems:*** AWS App Runner, AWS Bedrock (Nova Pro), Model Context Protocol (MCP), Playwright.
    - ***Impact:*** Built a self-directed research agent capable of navigating the live web to bypass LLM training cutoffs. Integrated the Model Context Protocol to securely connect a headless Playwright server, allowing the agent to autonomously scrape, read, and synthesize real-time financial news from javascript-heavy websites without manual intervention.
- **Cost-Optimized Serverless RAG Pipeline:**
    - ***Core Systems***: Amazon S3 (as Vector Store), Lambda, API Gateway, HuggingFace Embeddings.
    - ***Impact:*** Designed a high-performance "S3 Vectors" retrieval system that completely eliminated the need for expensive managed vector databases (e.g., Pinecone/OpenSearch). This architecture leveraged S3's low storage costs and Lambda's on-demand compute to achieve a 90% reduction in operational expenses for the document ingestion and retrieval subsystem while maintaining sub-second query latency.
- **Enterprise-Ready Serverless Infrastructure:**
    - ***Core Systems:*** Terraform, Amazon Aurora Serverless v2, CloudFront, AWS WAF.
    - ***Impact:*** Delivered a fully reproducible Infrastructure as Code (IaC) suite that provisions a secure, auto-scaling environment. Utilized Aurora Serverless v2 with the Data API to eliminate complex VPC networking overhead and leveraged "scale-to-zero" capabilities, ensuring the database incurs zero cost during idle periods—ideal for irregular workload patterns.
- **Full-Stack Product & Observability Layer:**
    - ***Core Systems:*** Next.js, Python FastAPI, Clerk Auth, LangFuse, CloudWatch.
    - ***Impact:*** Shipped a production-grade SaaS application that bridges complex AI backend logic with a responsive user interface. Implemented deep observability using LangFuse and CloudWatch dashboards to trace multi-turn agent execution paths, enabling rapid debugging of non-deterministic LLM behaviors and ensuring strict data isolation for user financial information.

SkyAgent: An LLM-Powered Autonomous Flight Booking System
Andela, *New York, USA*
https://github.com/hopeogbons/SkyAgent

- **Context-Aware LLM Orchestration Engine:**
    - ***Core Systems:*** OpenAI GPT-4o, Custom State Machine, Hybrid Regex/NLP Entity Extraction.

    - ***Impact:*** Architected a robust flight booking agent that handles complex multi-turn conversations and context switching (English/Pidgin). Implemented deterministic guardrails around the LLM to ensure 100% accurate extraction of critical booking parameters (dates, IATA codes) before triggering downstream APIs, preventing hallucinated transactions in production.
- **Distributed Real-Time Data Ingestion Pipeline:**
    - ***Core Systems:*** FastAPI Microservice, Redis Task Queue, Selenium Worker Pool.
    - ***Impact:*** Built a scalable, asynchronous scraping infrastructure to aggregate live pricing from legacy airline portals lacking public APIs. Deployed a concurrent worker pool with anti-detection measures that reduced data retrieval latency by ~60% compared to sequential processing, ensuring users receive up-to-the-second inventory.
- **Resilient Tool Execution & Fallback Strategy:**
    - ***Core Systems:*** Function Calling (Tool Use), External API Integration (Aviation Edge), Error Handling.
    - ***Impact:*** Designed a fail-safe execution layer that autonomously creates bookings and queries schedules. Implemented a hierarchical data retrieval strategy that attempts high-fidelity API sources first and gracefully degrades to scraping or web search (Serper), maintaining system availability even during partial provider outages.
- **Rapid Prototyping & Stakeholder Interaction Layer:**
    - ***Core Systems:*** Gradio, SQLite, Persistent Session Management.
    - ***Impact:*** Delivered a fully functional, end-to-end interface for immediate internal testing and stakeholder demos. This allowed for rapid iteration on prompt engineering and user experience (UX) flows without the overhead of a dedicated frontend team, accelerating the feedback loop by weeks.

Showcase: An MCP-Driven Agentic System for Automated Order Management

Andela, *New York, USA*

https://github.com/hopeogbons/showcase

- **Secure AI Customer Support Agent:**
    - ***Core System:*** OpenAI (GPT-4o), Model Context Protocol (MCP), Python, Gradio, AWS App Runner.
    - ***Impact***: Engineered a secure agentic workflow that bridges natural language intent with backend systems; implemented a custom middleware layer that automatically injects authenticated user contexts into tool calls, enabling the agent to autonomously managing orders while enforcing 100% data privacy and authorization compliance.
- **Full-Stack LLM Observability Pipeline:**
    - ***Core System:*** Langfuse, OpenTelemetry, OpenAI SDK.
    - ***Impact:*** Integrated end-to-end distributed tracing for all LLM generations and tool executions, enabling real-time monitoring of agent reliability; this reduced debugging time for hallucinations and provided a complete audit trail for compliance-sensitive actions.
- **Context-Aware Session Management System:**
    - ***Core System:*** Gradio, REST APIs, JSON-RPC.
    - ***Impact:*** Designed a stateful conversation architecture that synchronizes frontend authentication with dynamic system prompting; this captured and persisted user "sessions" across multi-turn dialogues, reducing time-to-resolution by allowing the agent to proactively reference active orders and history without re-prompting.
- **Rapid Prototyping & API Integration Strategy:**
    - ***Core System:*** Jupyter Notebooks, Requests, Dynamic Tool Mapping.

  - ***Impact:*** Developed a schema-driven discovery process to map existing REST endpoints to LLM-compatible tool definitions; this decoupled the frontend from backend logic, allowing rapid iteration of agent capabilities without requiring core application redeploys.

MediNotes Pro: AI-Powered Clinical Documentation Assistant

Andela, *New York, USA*

https://github.com/hopeogbons/medinotes-pro

- **Real-Time Clinical AI Assistant:**
  - ***Core Systems:*** Python (FastAPI), OpenAI GPT-5, Server-Sent Events (SSE), React (Next.js).
  - ***Impact:*** Engineered a low-latency generative AI pipeline that streams consultation summaries in real-time. Reduced physician documentation turnaround by 40% while maintaining a non-blocking, responsive user experience during high-load periods.
- **Secure SaaS Architecture & Monetization:**
  - ***Core Systems:*** Clerk Auth, Next.js Middleware, Role-Based Access Control (RBAC).
  - ***Impact:*** Architected a secure multi-tenant platform with seamless subscription gating. Implemented strict role-based access control to ensure HIPAA-ready data privacy, successfully automating the upgrade flow for premium medical features.
- **Unified Hybrid Deployment System:**
  - ***Core Systems:*** Docker (Multi-stage), AWS App Runner, Static Site Generation (SSG).
  - ***Impact:*** Designed a self-contained hybrid deployment strategy bundling a static frontend with a dynamic backend. Simplified infrastructure complexity and eliminated cross-origin latency, ensuring consistent, zero-downtime releases on cloud serverless platforms.
- **Context-Aware Prompt Engineering:**
  - ***Core Systems:*** Prompt Engineering, Context Windows, Structured Output.
  - ***Impact:*** Developed sophisticated system prompts to transform unstructured clinical notes into structured medical records. Achieved high fidelity in medical terminology and tone consistency, minimizing hallucinations across diverse patient scenarios.

FOUNDATIONAL ENGINEERING EXPERIENCE

Senior Software Engineer / Forward-Deployed AI Engineer

Andela Inc., *New York, America*

(May 2019 – Present)

Forward-Deployed Client Work

- ***Headspace Health***
  Led frontend architecture and feature delivery for the Headspace Care platform, designing and implementing in-app appointment scheduling workflows using React, Redux Toolkit, and GraphQL/RPC APIs; partnered closely with design to translate Figma mocks into responsive, production-grade UI, modernized legacy state management to improve data consistency and UX, and reduced defects through rigorous reviews and engineering standards.
- ***Medable, Inc.***
  Improved Medable’s Televisit application and associated tooling to support remote patient engagement across decentralized clinical trials; collaborated with cross-functional teams to build robust configuration pipelines, CLI utilities, and production-ready integration layers that improved developer velocity, enhanced clinical data quality, and ensured seamless interoperability with Medable’s cloud-based clinical trial platform.

- ***Medici Technologies, LLC***
  Built and enhanced Medici's Televisit platform and associated engineering tooling to support secure, HIPAA-compliant remote care workflows; improved developer workflows and ensured seamless backend interoperability, contributing to a robust telehealth experience for clinicians and patients.
- ***Groove Labs***
  Managed a cross-functional squad of 4 engineers, conducting deep-dive code reviews and establishing full-stack standards for the **React/Saga** and **Rails/Sidekiq** environments.

  Collaborated directly with Product and Revenue teams to translate complex VoIP and CRM requirements into scalable technical roadmaps that drove Groove's #1 ranking in Enterprise Customer Satisfaction.

Cross-Client Impact

- Embedded within client teams as a senior/principal-level engineer, owning design, implementation, and delivery of production systems.
- Partnered with stakeholders to translate ambiguous requirements into shipped software.
- Mentored engineers and set technical standards across multiple engagements.

Software Developer Coordinator

eHealth Africa, *Kano, Nigeria*
(February 2017 – March 2019)

Developed and deployed mission-critical, offline-capable health informatics platforms using **Python/Django, React, and Angular**. **Executed** high-impact engineering solutions for global stakeholders like the **Bill & Melinda Gates Foundation**, specializing in high-concurrency systems including the **EMID Vaccination Dashboard** and the **Kano Health Facilities Registry**. **Built** end-to-end technical infrastructure that converts complex donor requirements into real-world health outcomes, directly advancing healthcare delivery for millions across West Africa through enhanced data integrity and **GIS-integrated systems**.

Lead Software Developer / Independent Consultant

RoboOffice Ltd, *Lagos, Nigeria*
(April 2014 – December 2016)

Delivered mission-critical digital transformations for high-impact organizations, including an enterprise-scale ERP for **Redline Logistics** and a high-availability drone delivery system for **Zipline** using **React**, **Django**, and **Node.js**. **Architected** a bespoke PropTech payment platform for **Adron Homes** to manage multi-tier installment plans and **engineered** a specialized POS solution for **Imagine Boutique** to digitize retail operations. Across these portfolios, **modernized** core business workflows and eliminated administrative bottlenecks, driving a **35% increase in operational efficiency** through scalable, end-to-end technical infrastructure that converted manual processes into high-performance digital systems.

Software Developer

Syssoft Consultants Ltd, *Lagos, Nigeria*

(December 2010 – March 2014)

Modernized and maintained mission-critical Life and Non-Life insurance enterprise suites for tier-1 firms including **IGI, ARM, and NEM**, utilizing **PHP, MySQL, and JavaScript**. **Revamped** the entire visual architecture and frontend performance without breaking legacy dependencies, ensuring fluid data grid accessibility across diverse hardware. **Engineered** core updates to claims, policy management, and premium calculation modules while managing complex version control via **SVN Tortoise**. **Delivered** high-level technical consultancy and documentation that maintained **99.9% uptime** for premium clients, successfully translating intricate insurance requirements into scalable, production-ready software improvements..

EDUCATION

Bachelor of Science (BSc.) in Computing
University of Portsmouth, *UK*

Professional Diploma in Software Engineering: Java Technology
NIIT, *Lagos, Nigeria*

Diploma in Computer Technology
Obafemi Awolowo University, *Osun, Nigeria*

CERTIFICATIONS

- **OpenAI for Developers** - Pluralsight
- **Machine Learning Engineering** - Pluralsight
- **Large Language Models - Prompt Engineering** - Pluralsight
- **Retrieval-Argumented Generation (RAG)** - Pluralsight
- **AI Builder with n8n: Create Agents & Voice Agents** - *Udemy*
- **AI Leadership Track: Gen AI, Agentic AI for Business Leaders** - *Udemy*
- **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** - *Udemy*
- **AI Engineer Agentic Track: The Complete Agent & MCP Course** - *Udemy*
- **JavaScript Algorithms and Data Structures Masterclass** - *Udemy*
- **Modern React 18 Bootcamp** - *Udemy*
- **React Testing Library and Jest: The Complete Guide** - *Udemy*
- **Redux Saga** - *Pluralsight*
- **GraphQL by Example** - *Udemy*
- **Jira and Greenhopper for Agile Teams** - *Udemy*
- **Agile Methodologies** - *Callif Technologies*

REFERENCES
Referees & supporting documents are available upon request.